Training LLMs
Building SLMs From LLMs: Distilling, Shrinking, And Finetuning.

How to Make LLMs Smaller and Cheaper

- By Dr. Dakshinamurthy V Kolluru

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Reinforcement Learning

- Instead of copying examples, think of it like training a puppy: you reward good behavior (like sitting on command) and discourage bad ones. However, reward is sometimes difficult to specify as a formal function in conversations.

- RL as a field worked well for games, where reward was winning.

- RL with human feedback (RLHF) uses human feedback to determine the desired behavior, often by training a *reward model*.

  - P. Christiano, J. Leike, T. B. Brown, M. Martic, S. Legg, and D. Amodei, Deep reinforcement learning from human preferences, *NeurIPS*, 2017.
  - B. Ibarz, J. Leike, T. Pohlen, G. Irving, S. Legg, and D. Amodei, Reward learning from human preferences and demonstrations in Atari, *NeurIPS*, 2018.

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Why RL works?

- RL provides explicit negative feedback. Actions that lead to lower rewards are penalized. Although there is no credit assignment and all actions leading to a poor outcome are down weighted without fine-grained attribution, the model nonetheless receives direct signals about which behaviors to avoid.
- Studies show RLHF boosts user satisfaction and cuts toxicity (harmful content) by 50% or more (Askell et al., 2021). Without RL, models might be grammatically perfect but unengaging or risky. RL adds that extra polish, like turning a robot into a friendly companion.

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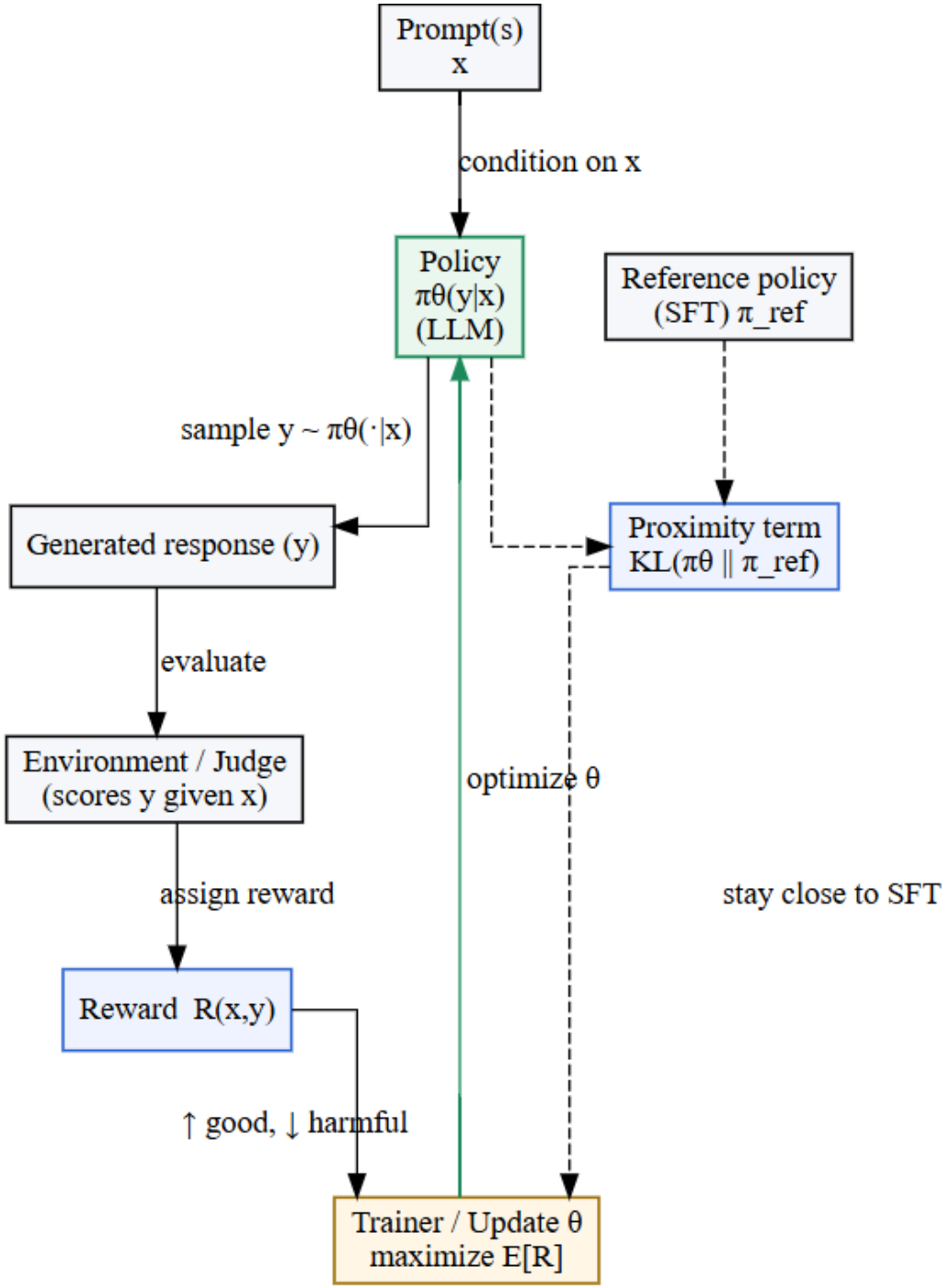
RL process



Verifiable rewards

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DeepSeek: RLVR

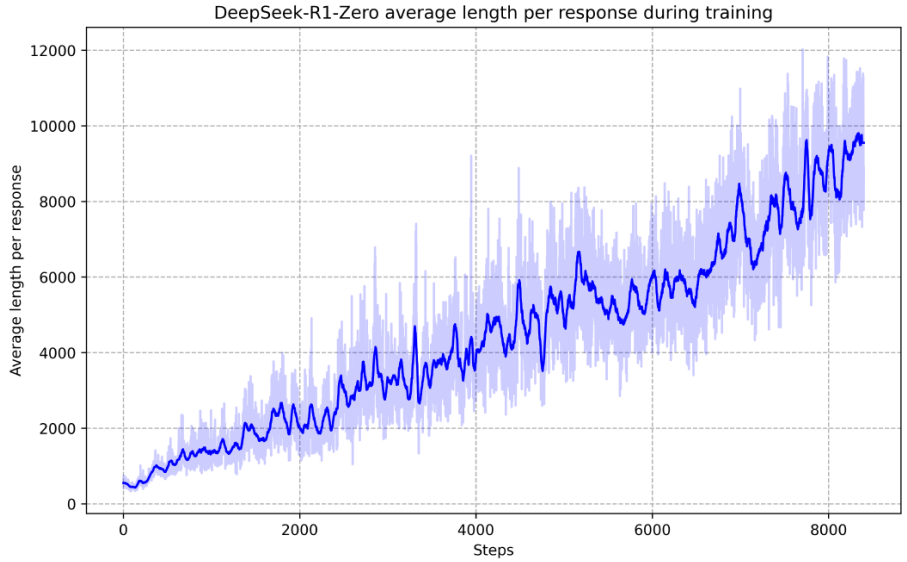

Figure 3 | The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time.

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$

$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$

Rearrange to isolate the inner square root term:

$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$

$\ldots$

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$

We started with the equation:

$\sqrt{a - \sqrt{a + x}} = x$

First, let's square both sides:

$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$

Next, I could square both sides again, treating the equation: $\cdots$

$\ldots$

These are learned by just looking at the correct answers

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RLVR and unexpected emergence

- “One of the most remarkable aspects of this self-evolution is the emergence of sophisticated behaviors as the test-time computation increases for models trained with RLVR
  - These behaviors are not explicitly programmed but instead emerge as a result of the model’s interaction with the reinforcement learning environment.”
- These worked for Qwen, Deepseek which were trained on RLVR and not Llamma or other models which were not trained!!

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RLVR works with cognitive behavior

- Qwen and DeepSeekbase models exhibit certain key cognitive behaviors for reasoning: verification (systematic error-checking), backtracking (abandoning failing approaches), subgoal setting (decomposing problems into manageable steps), backward chaining (reasoning from desired outcomes to initial inputs).

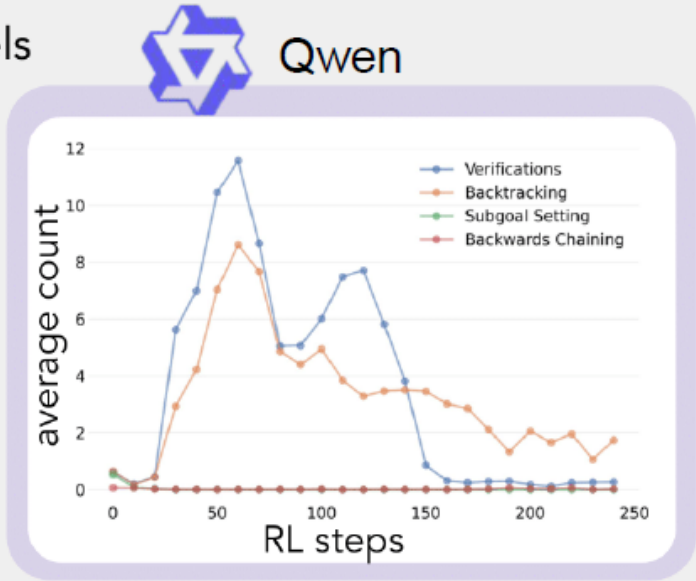
A contrast in behaviors explored by the two models

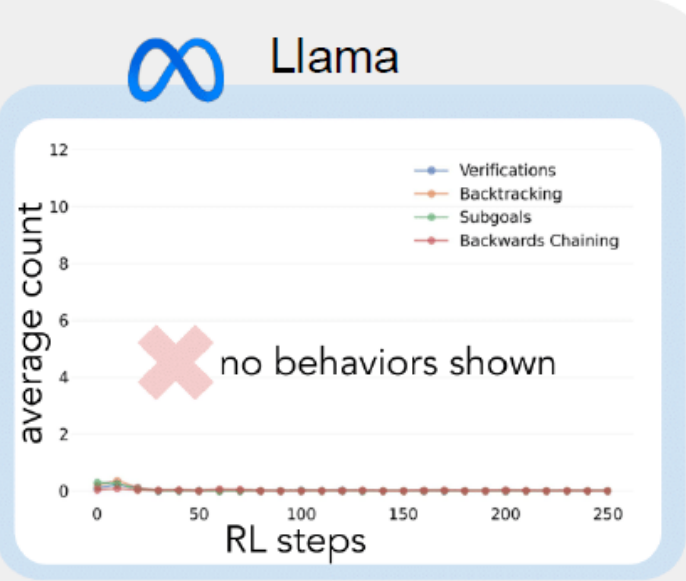
Verifications
"Let me check my answer …"

Subgoal Setting
"Let's try to get to a multiple of 10"

Backtracking
"Let's try a different approach, what if we …"

Backward Chaining
"Working backwards, 24 is 8 times 3"





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Fine-Tuning and PEFT (Parameter-Efficient Fine-Tuning)

When prompts aren't enough… teach the model new behavior

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Why Fine-tune An LLM?

- Prompting = surface-level control
- Fine-tuning = embed knowledge or style into the model
- Prompting (temporary fix) vs Fine-Tuning (permanent learning)

Start with prompting.

*Use RAG if the model needs **external knowledge**.*
Use fine-tuning when the model itself needs to change how it behaves.

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How do you add LORA

- 100X100 need 10,000 connections
- 100X2X100 need 400 connections

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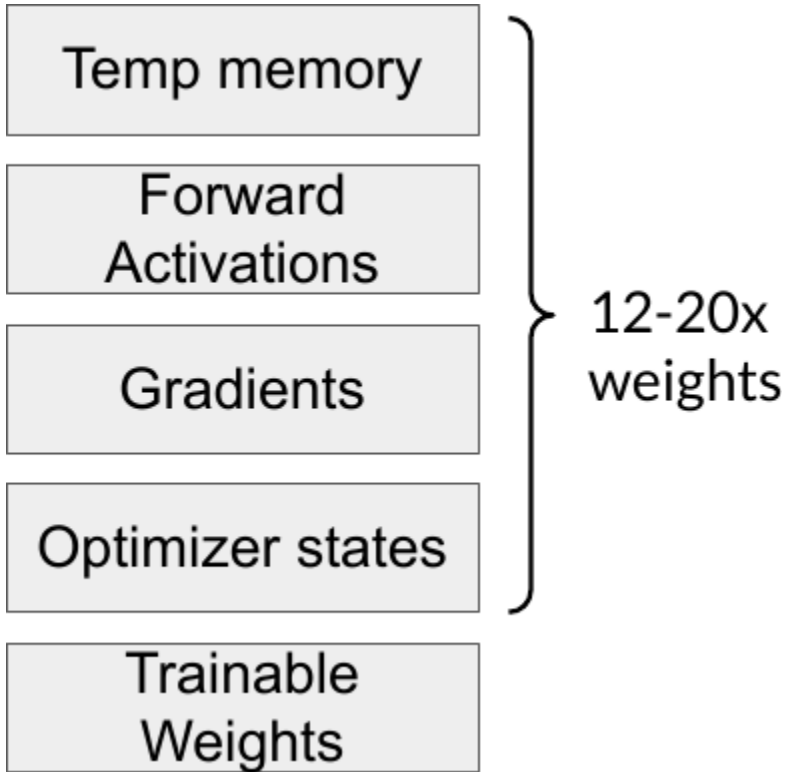
Full fine-tuning of large LLMs is challenging

- Full finetuning requires large number of computational resources.
- GPU memory requirement is equivalent to training the LLM from scratch.
- High budget implications (lots of samples, compute etc.)
- Need to create a large finetuned LLM per task or a group of tasks.
- Catastropic forgetting if the training is not extensive



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PEFT Methods

Reparameterization

Reparameterize model weights using a low-rank representation

LoRA

Additive

Add trainable layers or parameters to model

Adapters

Soft Prompts

Prompt Tuning

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LoRA: Low Rank Adaption of LLMs

Step-1: Freeze most of the weights of the LLM. For example, the attention weights and the FCN weights.

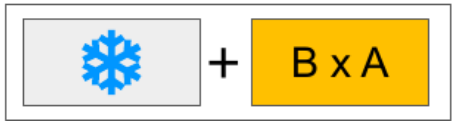
Step-2: introduce 2 rank matrices decomposition .

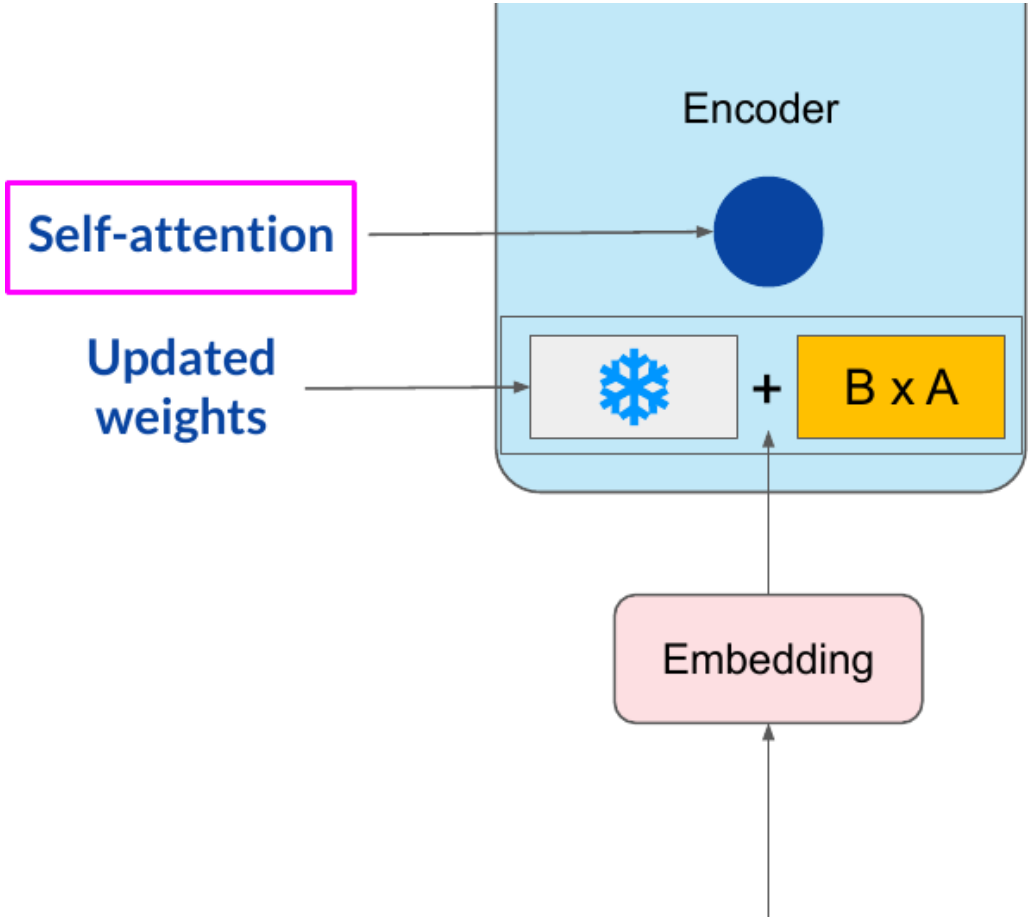
Step-3: Train the weights of the smaller matrices

Steps to update model for inference

1. Matrix multiply the low rank matrices

2. Add to original weights





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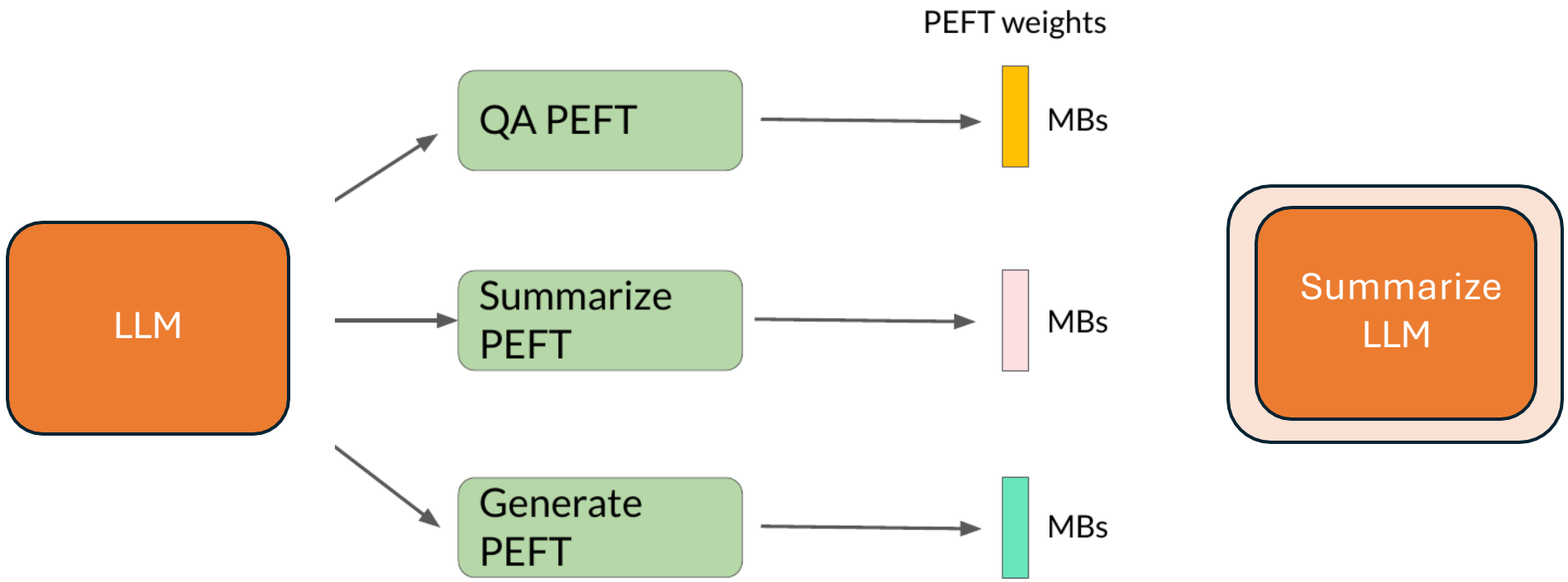
PEFT finetuning saves cost and storage



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Lora: Parameter Reduction Example

- Query matrix from BERT second encoder layer has dimensions d x k = 768 x 768
- So, 768 x 768 = 5,89,824 trainable parameters
- In LoRA with rank r = 32:
- A has dimensions r x k = 32 x 768 = 9216 parameters
- B has dimension d x r = 768 x 32 = 9216 trainable parameters
- 92% reduction in parameters to train!

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Quantization

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Smaller LLMs: Quantization

Quantization (Float → Int):

$$q = \text{round}\left(\frac{x}{\text{scale}}\right) + \text{zero_point}$$

Dequantization (Int → Float):

$$x \approx (q - \text{zero_point}) \times \text{scale}$$

Scale:

$$\text{scale} = \frac{x_{\max} - x_{\min}}{q_{\max} - q_{\min}} = \frac{x_{\max} - x_{\min}}{255}$$

Zero point:

$$\text{zero_point} = \text{round}\left(-\frac{x_{\min}}{\text{scale}}\right)$$

You may then clip or clamp `zero_point` into the valid quantization range.

Term	Meaning
x	Original float value
q	Quantized integer
scale	Floating-point step size between quantized levels
zero_point	Integer that maps float zero to a quantized value (usually in int range)

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Quantization

Quantization helps reduce the memory footprint of a model by storing numbers in 8-bit or 4-bit format instead of 32-bit floats.

For example, a 7B parameter model at FP32 takes ~28 GB; INT4 brings it down to ~3.5–4 GB. The tradeoff is slightly lower accuracy, but often acceptable.

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QLoRA = Quantized Model + LoRA Adapters

- QLoRA uses **4-bit quantized weights** for the base LLM → drastically reduces memory.
- Adds **LoRA adapters** for efficient fine-tuning → no need to update full weights.
- LoRA adapters are trained in **full precision**; only base model is quantized.
- Enables training of models like **LLaMA-7B on a single 24GB GPU**.
- Matches or exceeds full fine-tuning performance — at a fraction of cost and resources.

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Distillation

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Model Distillation

- A **“student model”** learns to mimic the outputs of a larger **“teacher model”**.
- Transfers knowledge from large models to smaller ones — fewer parameters, faster inference.
- Often used to compress models for real-time or mobile deployment.
- The student model learns from both:
- Ground truth labels
- Teacher’s logits or soft outputs (which carry richer info)
- Can preserve accuracy surprisingly well.

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Distilled Model Performance

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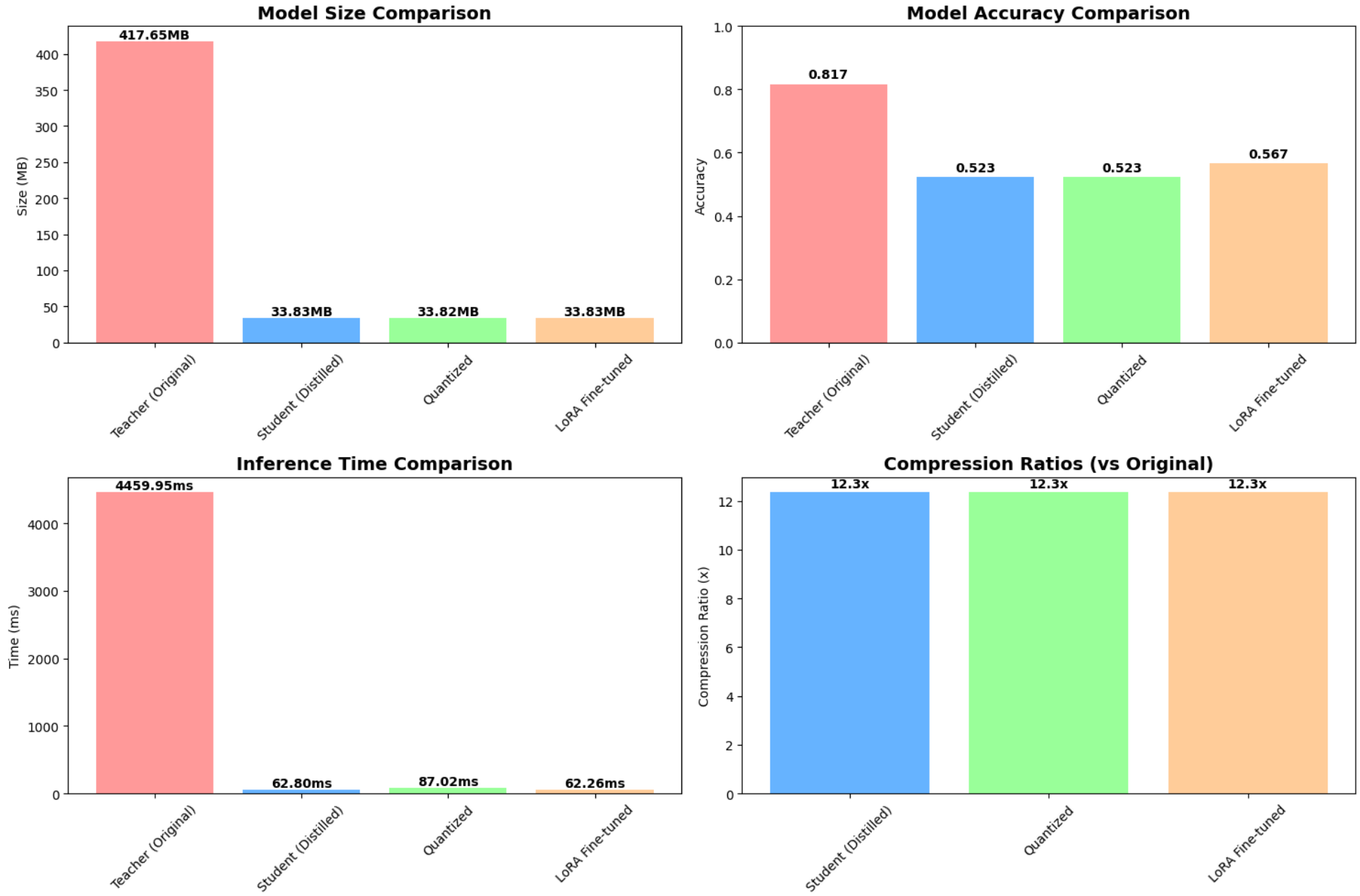

DistilBERT (66M) + 8-bit quantization → ~15MB model

Used in **mobile NLP apps** (e.g., **Snips Voice Assistant**) for fast, offline inference with minimal accuracy loss.

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Decision Matrix

When to use what

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Distillation vs Quantization

Feature	**Distillation**	**Quantization**
What it does	Trains a small model to mimic a large model	Reduces weight precision (e.g., FP32 → INT4)
Model Architecture	Student model is separate from teacher	Model architecture remains the same
Training Needed	✅ Yes – Student needs to be trained	❌ No (Post-training quantization is common)
Accuracy Impact	Moderate loss (can be minimized)	Slight accuracy drop depending on bits used
Use Case	Create fast, lightweight inference models	Deploy large models on small hardware
Combines With	PEFT, task-specific tuning	LoRA (→ QLoRA), on-device models

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Choosing the right model optimization technique

Scenario / Need	Best Method	Why
Want lowest latency + smallest memory footprint	**Quantization (4-bit / 8-bit)**	Massive RAM reduction, deploy on low-cost GPUs/edge devices
Want small model that behaves like a big model	**Distillation**	Student captures teacher behavior; fast inference
Want to add *new behavior*, style, or output patterns	**PEFT / LoRA**	Efficient weight updates, minimal compute, preserves base LLM
Want to fine-tune *quantized model* on limited GPU (e.g., 24GB)	**QLoRA**	Full finetuning performance on 4-bit model
Need domain-context answers, but not new model behavior	**RAG (next module)**	External knowledge → no model changes
Need full weights updated for highly specialized tasks	**Full Finetuning**	Maximum flexibility, but highest compute/storage cost

Most enterprise tasks can be solved with Quantization + LoRA / QLoRA. Full finetuning is rarely needed today

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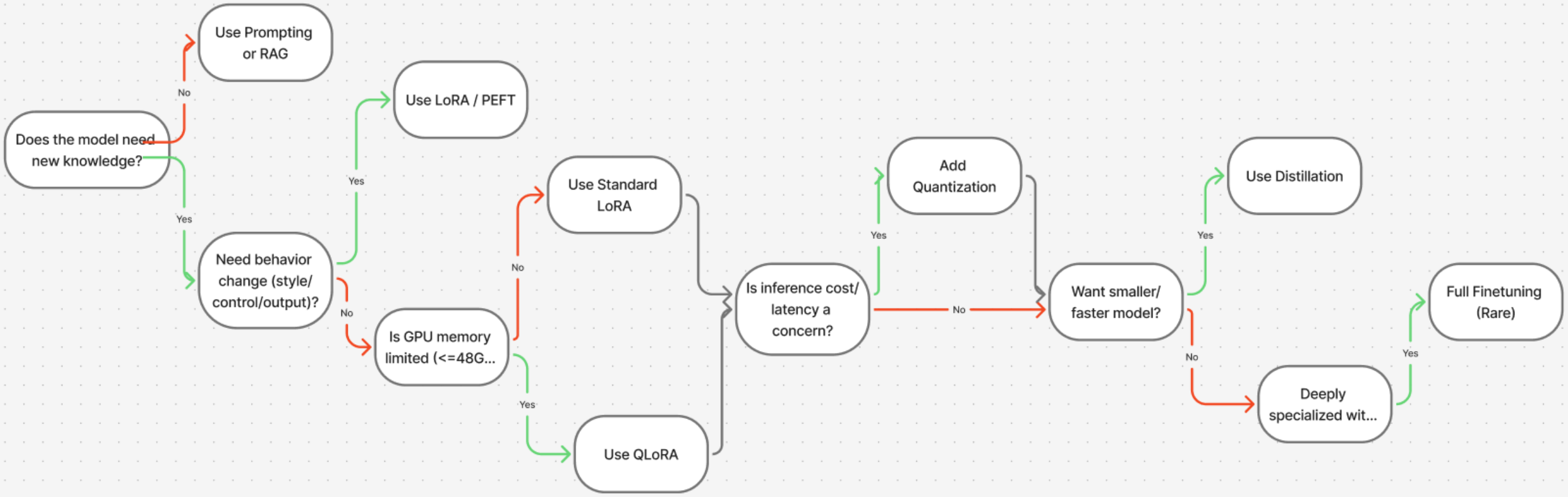
Practical Decision Flow for Enterprises



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Thank You

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